

Vessel MAPS Template (Instrumentation)

The Vessel MAPS template is designed to control a vessel that is connected to digital and/or analog level control units.

All MAPS templates provide the following:

- an associated PLC function block
- a set of SCADA tags (Adroit agents) linked to the provided signals
- an associated SCADA graphic with inbuilt faceplates

This MAPS template is available in three models – advanced, standard and basic, which provide respectively lower levels of control over the vessel.

This document describes the Advanced Vessel MAPS Template and also defines the level of functionality provided by the standard and basic models. So that by reading this document you can decide which model best suits your needs. You should consider the following factors in your choice:

- the importance of the vessel being controlled and
- the budget of the customer

Since the more complex the model, the greater the required PLC memory usage; number of PLC I/O and SCADA size (scan points) - see the **Resources** table below.

Note: The version 2 (v2) Vessel (Advanced and Standard) MAPS template, communicates Floating point values instead of Integer values to the driver.

Advanced Vessel Functional Philosophy

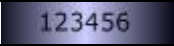
The vessel is filled according to the level field inputs received.

Field inputs

- High-high Level
- High Level
- Low Level
- Low-low Level
- Analog level Input

Simulation mode - will override inputs to the vessel and simulate a level in the PLC program that is controlled from the faceplate. Simulation mode is used to test the process logic without receiving the physical input itself. It is very useful in testing, commissioning and maintenance.

Typical Faceplate Graphics for the Advanced Vessel MAPS Template

Vessel 	
	Level Indication (High-high, High, Low and Low-Low)
	Analog level Indication Value
	Control / Tag Name
Graphics legend:	
Fill colour	Vessel Status
Grey	Empty Level
Blue	Analog level fill indication

Signal Description		Agent Type	Advanced	Standard	Basic
SCADA Control Words:					
SSCL 0	Control Word	Marshal	x	x	x
SSCL 1	High-high Level	Analog	x	x	
SSCL 2	High Level	Analog	x	x	
SSCL 3	Low Level	Analog	x	x	
SSCL 4	Low-low Level	Analog	x	x	
SSCL 5	Scale Min In Setpoint (Raw Min)	Analog	x	x	
SSCL 6	Scale Max In Setpoint (Raw Max)	Analog	x	x	
SSCL 7	Scale Min Out Setpoint (Eng. Min)	Analog	x	x	
SSCL 8	Scale Max Out Setpoint (Eng. Max)	Analog	x	x	
SSCL 9	Simulation Value	Analog	x	x	
SSCL 10	Alarm Hysteresis	Analog	x	x	

Signal Description		Agent Type	Advanced	Standard	Basic
SCADA Status Words:					
SSSL 0	Status Word	Marshal	x	x	x
SSSL 1	Vessel Process Value (Scaled Value)	Analog	x	x	
SSSL 2	Vessel RAW Value (Analog Raw Value)	Analog	x	x	

Digital Inputs:		Advanced	Standard	Basic
DI 0	High-high Level	x		x
DI 1	High Level	x		x
DI 2	Low Level	x		x
DI 3	Low-low Level	x		x
Analog Input:				
AI 0	Analog Input for Vessel Level	x	x	

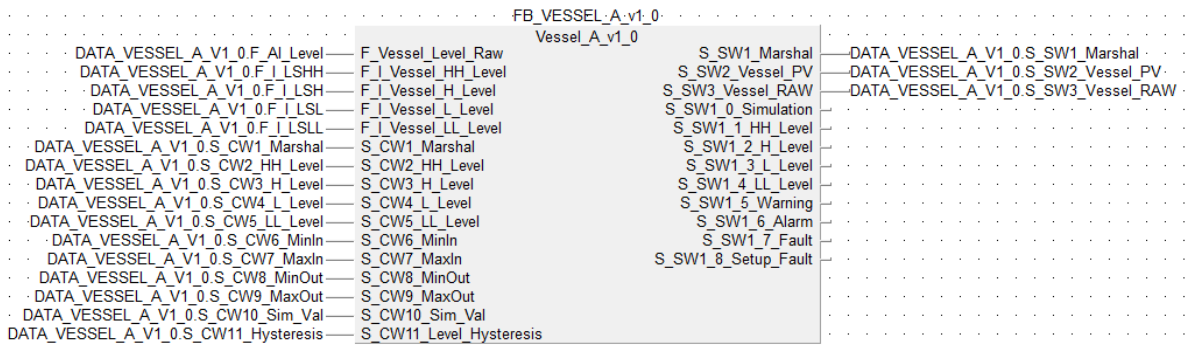
SCADA Control Word (Adroit Marshal Agent)		Event Logged	Advanced	Standard	Basic
Bit 0	Simulation	x	x	x	x
Bit 1	High-high AI Level Enabled	x	x	x	
Bit 2	High AI Level Enabled	x	x	x	
Bit 3	Low AI Level Enabled	x	x	x	
Bit 4	Low Low AI Level Enabled	x	x	x	
Bit 5	Reset Fault		x	x	x
Bit 6	High-high DI Level Enabled	x	x		x
Bit 7	High DI Level Enabled	x	x		x
Bit 8	Low DI Level Enabled	x	x		x
Bit 9	Low-low DI Level Enabled	x	x		x
Bit 10	Spare				
Bit 11	Spare				
Bit 12	Spare				
Bit 13	Spare				
Bit 14	Spare				
Bit 15	Spare				

SCADA Status Word (Adroit Marshal Agent)		Alarmed	Advanced	Standard	Basic
Bit 0	Simulation On		x	x	x
Bit 1	High-high Level Active	x	x	x	x
Bit 2	High Level Active	x	x	x	x
Bit 3	Low Level Active	x	x	x	x
Bit 4	Low-low Level Active	x	x	x	x
Bit 5	Level Warning	x	x	x	x
Bit 6	Level Alarm	x	x	x	x
Bit 7	Level Fault Latch		x	x	x
Bit 8	Level Setup Fault		x	x	
Bit 9	Spare				
Bit 10	Spare				
Bit 11	Spare				
Bit 12	Spare				
Bit 13	Spare				
Bit 14	Spare				
Bit 15	Spare				

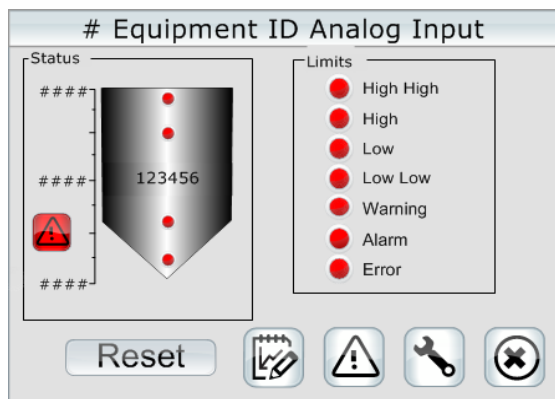
Resources	Advanced	Standard	Basic
Digital Inputs	4	0	4
Digital Outputs	0	0	0
Analog Inputs	1	1	0
Analog Outputs	0	0	0
PLC Memory Steps	384	366	135
System Words Used	40	40	0
System Bits Used	76	72	62
SCADA SCAN Points	14	14	2

Template: Advanced Vessel PLC Function Block

<Device> (Advanced Vessel Control)

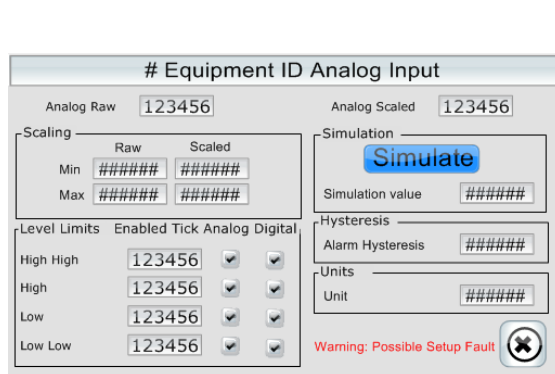


Template Graphic: Advanced Home Screen

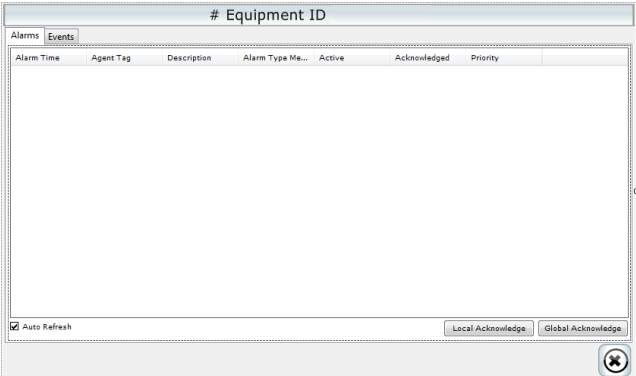


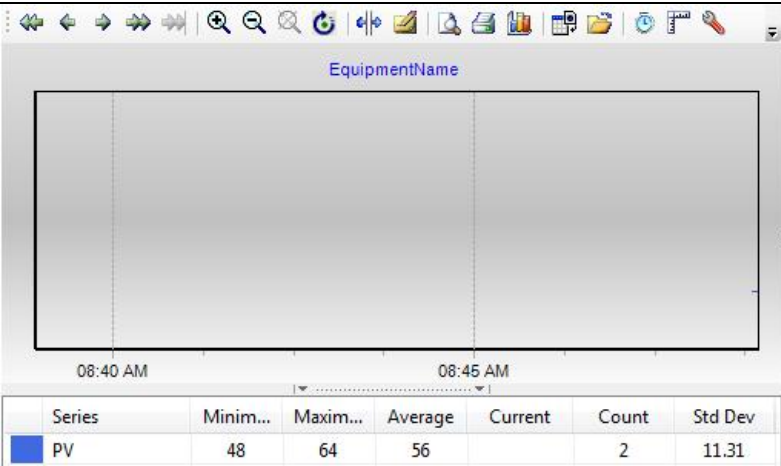
Control	Descriptions
# Equipment ID	Vessel controlled name
High High	High-high alarm indication
High	High warning indication
Low	Low warning indication
Low Low	Low-low alarm indication
Warning	Low, high alarm warning
Alarm	Low-low, high-high alarm fault
Error	Alarm fault latch
Reset	Reset the vessel after a fault has been repaired
123456	Analog scaled value indication
	Digital level indication
	Alarm fault latch

Template Graphic: Advanced Setup Screen



Control	Descriptions
Simulate	Enable simulation of vessel, disables the physical output
Simulation Value	Simulation value of vessel
Analog Raw	Raw value from field level
Analog Scaled	Scaled value from field level
Min, Max Raw	Raw level input values for scaling
Min, Max Scaled	Engineering level values for scaling
High High	High-high alarm setup value and enabling analog and Digital Inputs
High	High alarm setup value and enabling analog and Digital Inputs
Low	Low alarm setup value and enabling analog and Digital Inputs
Low Low	Low-low alarm setup value and enabling analog and Digital Inputs
Alarm Hysteresis	High and low analog level auto reset hysteresis value
Unit	Unit displayed for the scaled level of the vessel

Template Graphic: Advanced Alarms & Events Screen	Control	Descriptions
	Alarms	Display all the alarms filtered for the current Digital Output
	Events	Display all the events filtered for the current Digital Output.
	Local Acknowledge	Acknowledge all the alarms on locally on the current machine
	Global Acknowledge	Acknowledge all the alarms on globally on all the machines.

Template Graphic: Advanced Trend Screen	Control	Descriptions
	Equipment Name	Display the name of the current vessel of the trend
	PV	Display the level of the current level